

OT GoalAssist

"Helping Occupational Therapists COAST Through the Day."

Feasibility Presentation

CS 410: Professional Workforce Development

Old Dominion University

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Team Iron

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Meet Team Iron!



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The Motivation: Why Occupational Therapy?

Helping individuals develop, recover, and maintain the skills they need to participate in everyday activities.

Children



Development and participation

Older Adults



Safety and independence

Stroke & Injury Survivors



Regaining everyday skills

169,600

Occupational therapist jobs in 2025 alone

15%

Projected employment growth for occupational therapists through 2035.

Sources: U.S. Bureau of Labor Statistics.

How does OT Care Work?

01
Patient is tested
Assess abilities, limitations, and functional needs

02
Care Plan
Therapist creates an individualized treatment plan

03
COAST Goals
Write measurable, functional goals

04
Insurance Review
Review goals and medical necessity for coverage

Andrew's COAST Goal

C — Client

Andrew, a 7-year-old boy

O — Occupation

Trying new foods

A — Assistance

With minimal verbal encouragement

S — Specific Condition

During a structured feeding activity

T — Timeline

Within 8 weeks

Root Cause: Why Does the Problem Persist?

Occupational therapists are expected to do more with less time.

1. Productivity pressure
20.5% of 366 rehabilitation practitioners met burnout criteria.

Coleman et al., 2024

2. Insurance demands
Aetna and Cigna require specific, quantitative goals and documented progress.

Aetna; Cigna policies

The documentation bottleneck
Assessment scores + clinical observations + functional needs

Manual, time-consuming goal development
Therapists must translate clinical information into individualized, measurable goals while maintaining patient-care productivity.

Financial and professional consequences
Missing long-term goals and inadequate measurable goals are documented reasons for Medicare therapy claim denials.

First Coast Service Options, 2026

The Gap: Why Haven't Existing Tools Fixed This?

Rigid Dropdowns
Limit customization and flexibility

Generic templates
Fail to capture individualized patient needs

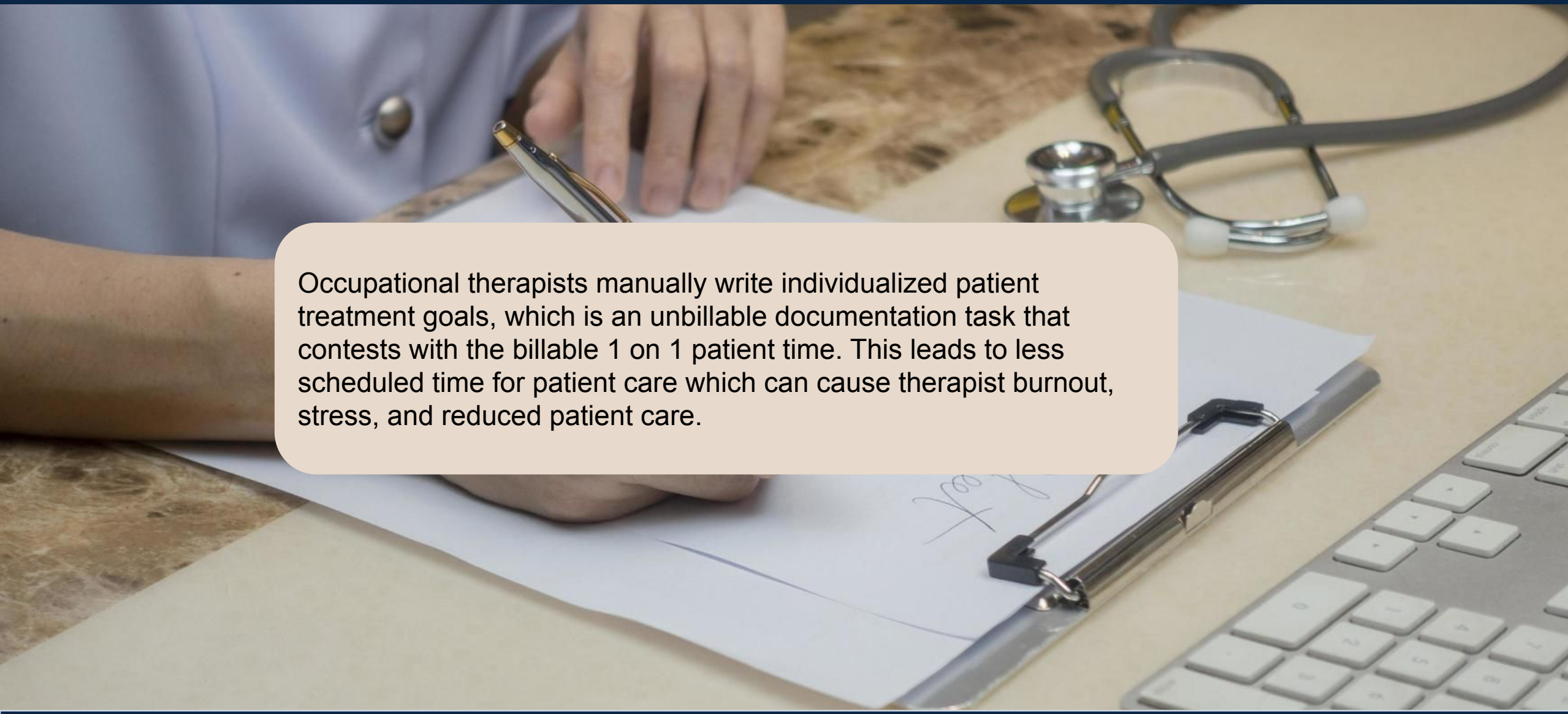
Blank text boxes
Require extensive manual typing

Diverse settings
Different populations and clinical requirements

Consequences

- Less time for patient care
- High burnout
- High staff turnover

Problem Statement



Occupational therapists manually write individualized patient treatment goals, which is an unbillable documentation task that contests with the billable 1 on 1 patient time. This leads to less scheduled time for patient care which can cause therapist burnout, stress, and reduced patient care.

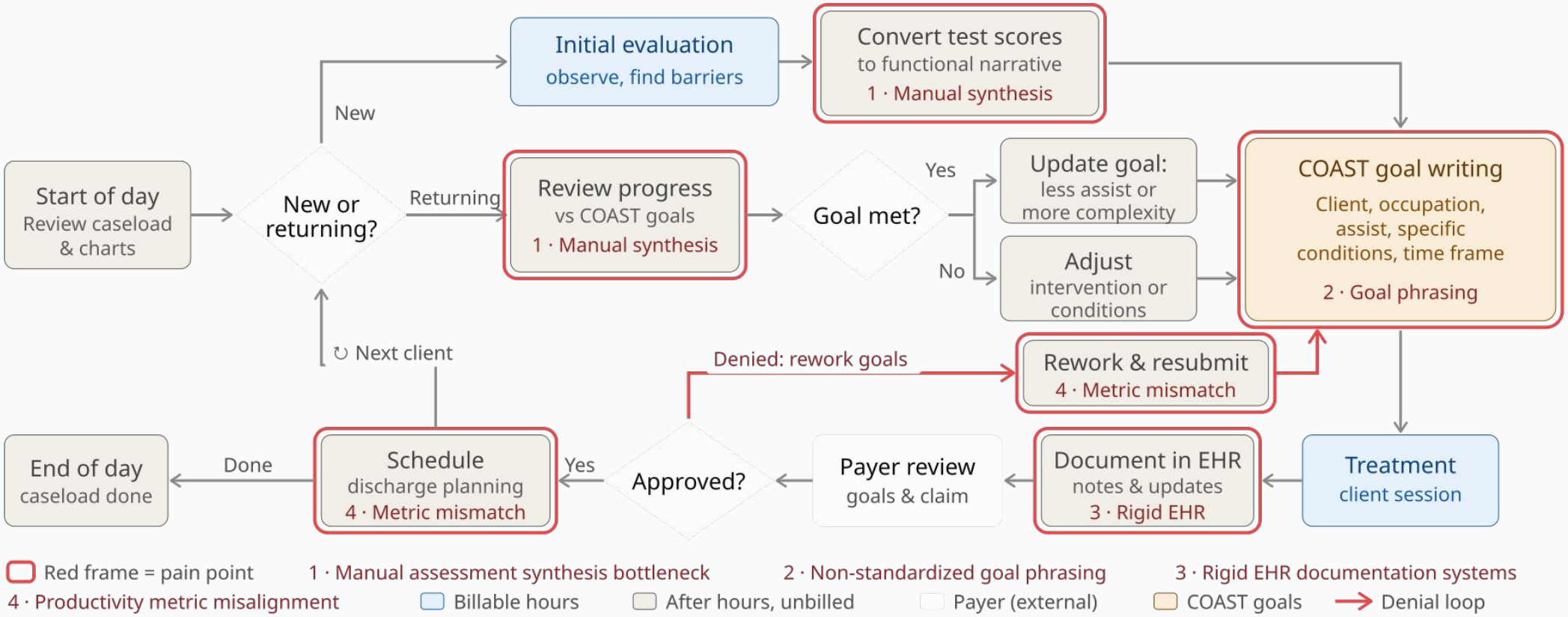
Target Ecosystem: Users, Customers, and Stakeholders

Direct Users	Customers	Stakeholders
Clinicians using the tool daily	*Organizations purchasing software*	*Beneficiaries & regulatory bodies*
• Registered Occupational Therapists (OTRs) • Certified Occupational Therapy Assistants (COTAs) • Core Pain Points: Lost personal evenings to paperwork ("pajama time"), mental fatigue from synthesizing test batteries, fear of insurance audits.	• Outpatient Therapy Private Practices • Hospital Rehabilitation Networks • School Districts & Pediatric Centers • Business Value: Maximizes billable treatment compliance, reduces clinician turnover, and prevents costly claim denials.	• Patients & Families: Receive clearer goals and faster progress tracking. • Insurance Payers (Medicare / Medicaid): Require objective, measurable proof of medical necessity. • Professional Bodies (AOTA): Advocates for clinician wellness and standards.

Problem Characteristics: Root Causes of the Documentation Gap

Identified Problem Characteristic (Systemic Gap)	Root Cause & Clinical Impact Description	Impact
1. Manual Assessment Synthesis Bottleneck	Standardized assessment batteries (BOT-2, PDMS-2, MMT) yield raw numerical percentiles. After deciding which areas to target with the resulting test data they write out each goal by hand for each patient. No one tool connects the scored results to the goal writing process, so OT's have to repeat this process every time.	Repetitive work
2. Missing COAST validation	Nothing in UI flags for missing COAST elements (such as explicit assistance levels or measurable environmental conditions). This triggers insurance claim rejections and audit delays.	Missing vital patient information, insurance claim rejections
3. Rigid EHR Documentation Systems	Current clinical software (e.g., WebPT, Epic) provides either rigid drop-down menus that fail to capture nuanced patient needs, or completely blank text boxes that force manual typing from scratch.	Non specific prompts/bad UI
4. Productivity Metric Misalignment	Productivity is measured based on billable patient treatment minutes. Drafting individual treatment goals is de-incentivized since this takes time away from the billable patient treatment minutes.	Documentation time takes up patient time

Current Process Flow: Day In The Life



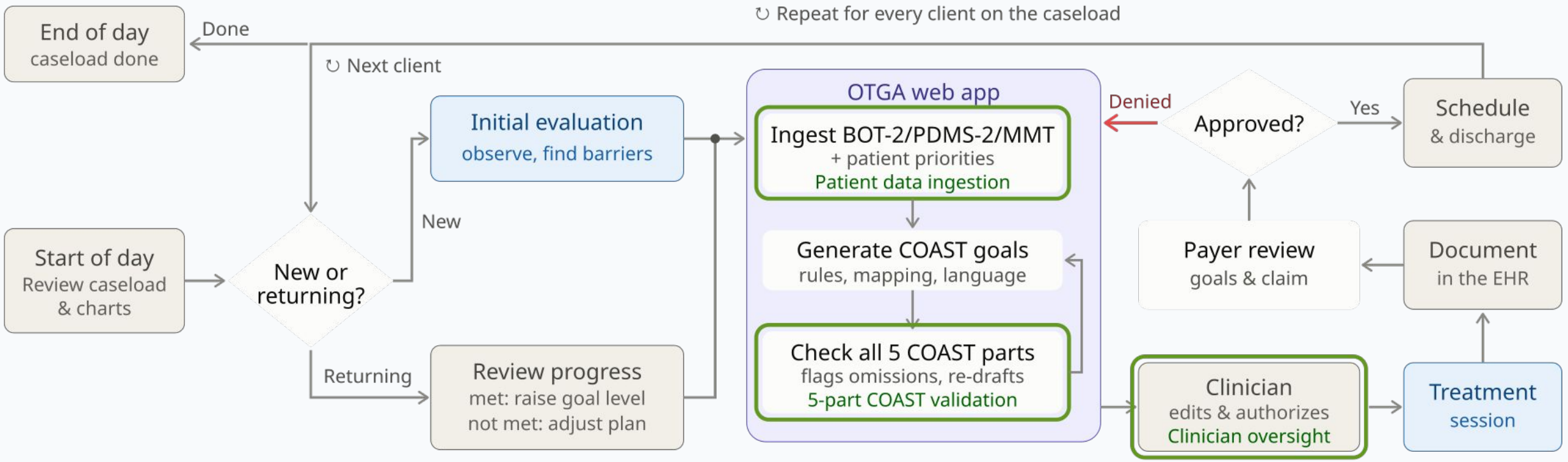
Core Takeaway: Goal-writing is a repetitive, non-billable cognitive overhead that pulls clinicians away from direct patient care. OT GoalAssist targets steps within the COAST timeline *(yellow highlight)* for intelligent automation. Pain points *(red)* make the process even more difficult combined with denial loops.

Proposed Solution: OT GoalAssist

OT GoalAssist is an intelligent web application designed for occupational therapy clinicians that accelerates the writing of individualized, compliant treatment goals. By ingesting standardized clinical assessment results and patient functional priorities, OT GoalAssist leverages a rule-based clinical engine, relational data mapping, and language synthesis to produce editable, COAST-compliant goal suggestions with automated omission detection. OT GoalAssist preserves clinician oversight while reducing the repetitive administrative workload.

Clinician Oversight	5-Part COAST Goal Validation	Patient Data Ingestion
Preserves full professional autonomy. The system provides structured recommendations that the licensed clinician inspects, edits, and authorizes under their clinical license.	Enforces complete five-part COAST syntax (Client, Occupation, Assist Level, Specific Condition, Timeframe), eliminating omission errors and reducing insurance claim denials.	Quick ingestion of standardized test results (BOT-2, PDMS-2, MMT) and patient records into functional goal statements.

Solution Process Flow: How It Works



Solution Characteristics

Problem Characteristic	Solution Characteristic	Solution
Manual Data Synthesis Bottleneck	**Automated Assessment Ingestion**	Quick-entry assessment forms (BOT-2, PDMS-2, MMT). For example, an OT could administer the BOT-2 using the official materials and calculate the results using an authorized scoring method. Then they will enter: the composite scores, percentile rank, age equivalent, and relevant subtest results. OT GoalAssist will then use those results to generate clinical observations and suggest treatment goals.
Non-Standardized Goal Phrasing	**COAST Validation**	Parser that validates the presence of Client, Occupation, Assist, Condition, Time in the goal.
Rigid EHR Documentation Systems	**Modular, Editable Interface**	Interactive interface allowing for the OT to review, modify, and export the goal in a user-friendly manner.
Productivity Metric Misalignment	**Accelerated Draft Synthesis**	Generates initial care plan drafts that includes treatment ideas as well as the completed COAST goal. The accelerated goal generation will free up OT time and allow for more time spent on patient care.

What OT GoalAssist Will Do

Feature Capability	What the Clinician Experiences & System Output
Structured Clinical Assessment Entry	Provides rapid-entry forms for standardized motor, sensory, and strength assessment scores (BOT-2, PDMS-2, MMT) alongside patient functional priorities.
Rule-Based COAST Goal Formulation	Translates numerical score deficits into clear, individualized 5-part COAST sentences using deterministic clinical phrasing rules.
Real-Time Omission Guardrails	Automatically highlights missing parameters (such as unspecified assistance levels or missing milestones) before care plans can be finalized.
Human-in-the-Loop Clinical Control	Gives clinicians full in-line editing controls, allowing real-time wording adjustments, goal deletion, or alternative phrase generation.
Universal EHR Clipboard Export	Formats approved goals into clean text blocks ready for instant copy-pasting into any existing clinical EHR system (WebPT, Epic, SimplePractice).

What OT GoalAssist Will NOT Do

OT GoalAssist will NOT	Why?
Replace any jobs	OT GoalAssist is simply a tool to help therapists do their job more efficiently by having a dedicated goal and note taker to track the progress and important contexts of individual patients. OT GoalAssist will only assist in keeping track with patients, not treating them. Therapists will still need to authorize goals with their license as liability.
Provide any medical advice or tell therapists how to do their job	OT GoalAssist is a tool to help a therapist keep track of their patients progress, situation and other important contexts or changes regarding them. It will not give therapists any information about how to treat patients or proceed with treatment.
Directly keep any patient information	OT GoalAssist is not a tool for patients and will not keep any information regarding them besides the relevant information that is explicitly provided to the therapist.

Technical Feasibility Snapshot: Architecture Overview

4-TIER POINT-OF-CARE ARCHITECTURE PIPELINE

SUB-SECOND LATENCY

TIER 1

Client Touch

Next.js 15 • PWA • iPad

Touch deficit intake & 1-click clipboard bridge directly at clinical point of care.

TIER 2

Processing Core

FastAPI • Python 3.12

Async memory pipeline with zero unencrypted PHI written to disk or public AI.

TIER 3

COAST Synthesizer

Deterministic Grammar

Rule-based slot filling guaranteeing all 5 COAST reimbursement criteria.

TIER 4

Compliant Vault

PostgreSQL • Supabase

Row-Level Security (RLS) & AES-256 encryption ensuring HIPAA compliance.

< 1.42s

Synthesis Latency

End-to-end template generation

100%

COAST Compliance

Zero missing reimbursement slots

0 PHI

Unencrypted Cloud Exposure

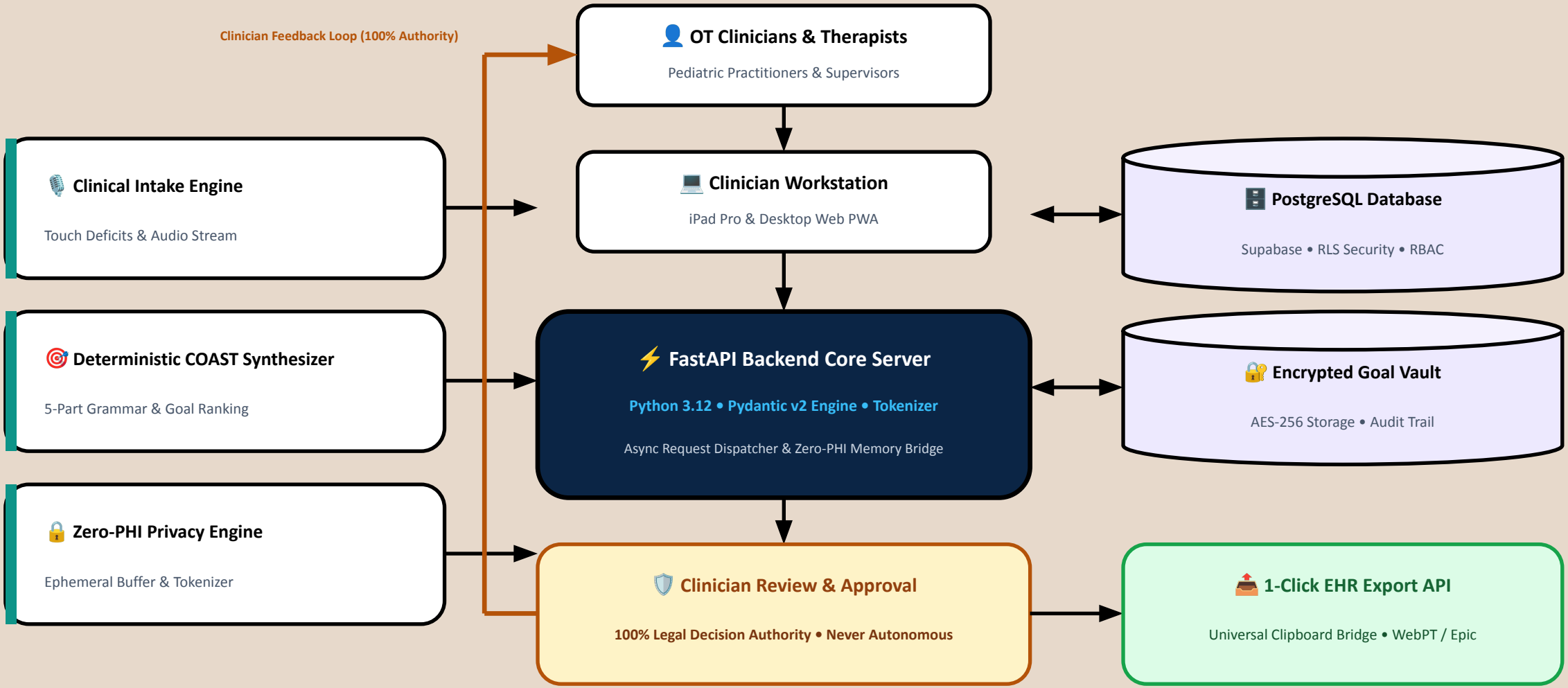
Zero PHI stored or sent to public LLMs

< 60s

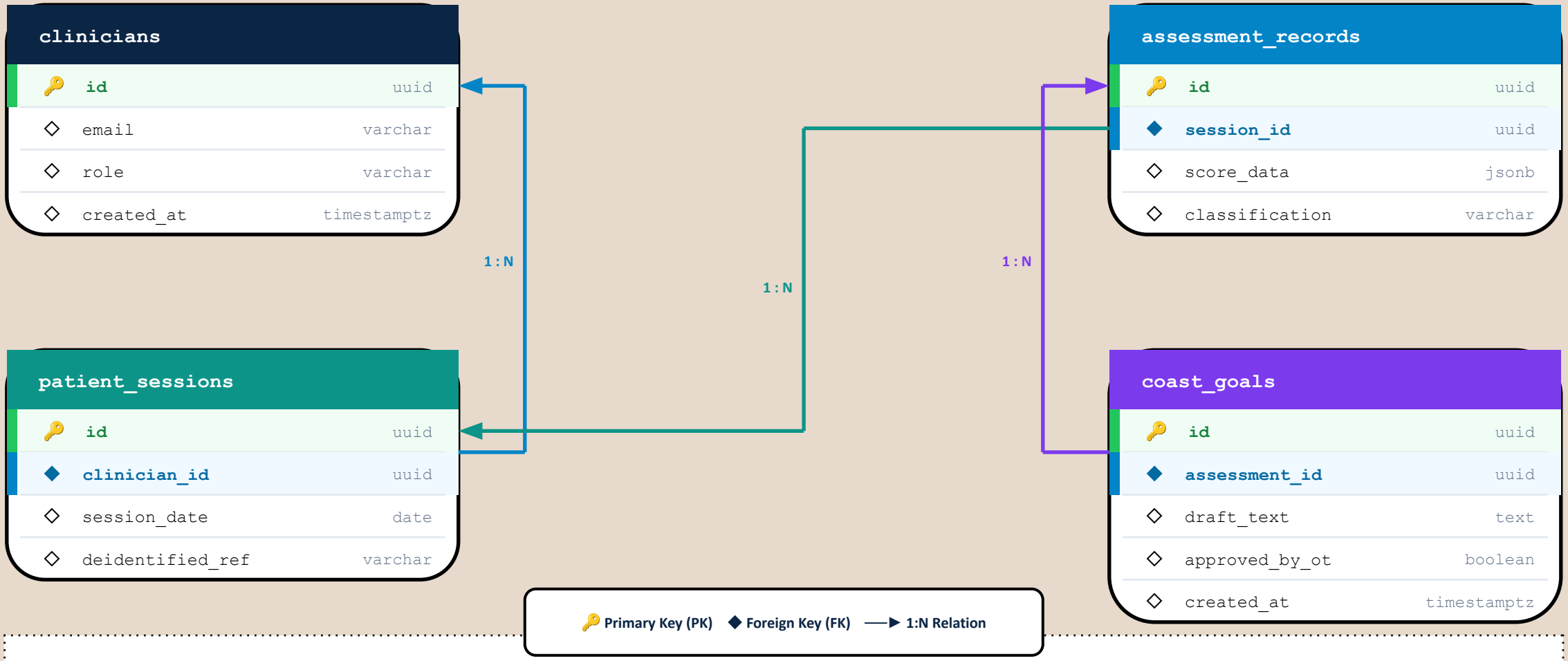
Point-of-Care Workflow

Recapturing 1.5-2.5h unbilled time

Major Functional Components Diagram (MFCD)



Relational Database Schema & Data Flow (ERD)



Core Database Architecture Principles: (1) Client-Side De-Identification strips patient names and MRNs before database transmission. (2) Ephemeral Session Records purge automatically upon export. (3) Strict Access Controls restrict data retrieval solely to the authenticated licensed therapist.

Development Tools & Implementation Environment

- **Frontend Dev Tools**
  - HTML5
  - Modern CSS
  - JavaScript
  - TypeScript
- **Backend Dev Tools**
  - Node.js
  - Java
  - Python
- **Continuous Integration and Deployment**
  - Git
  - GitHub
  - GitHub Pages



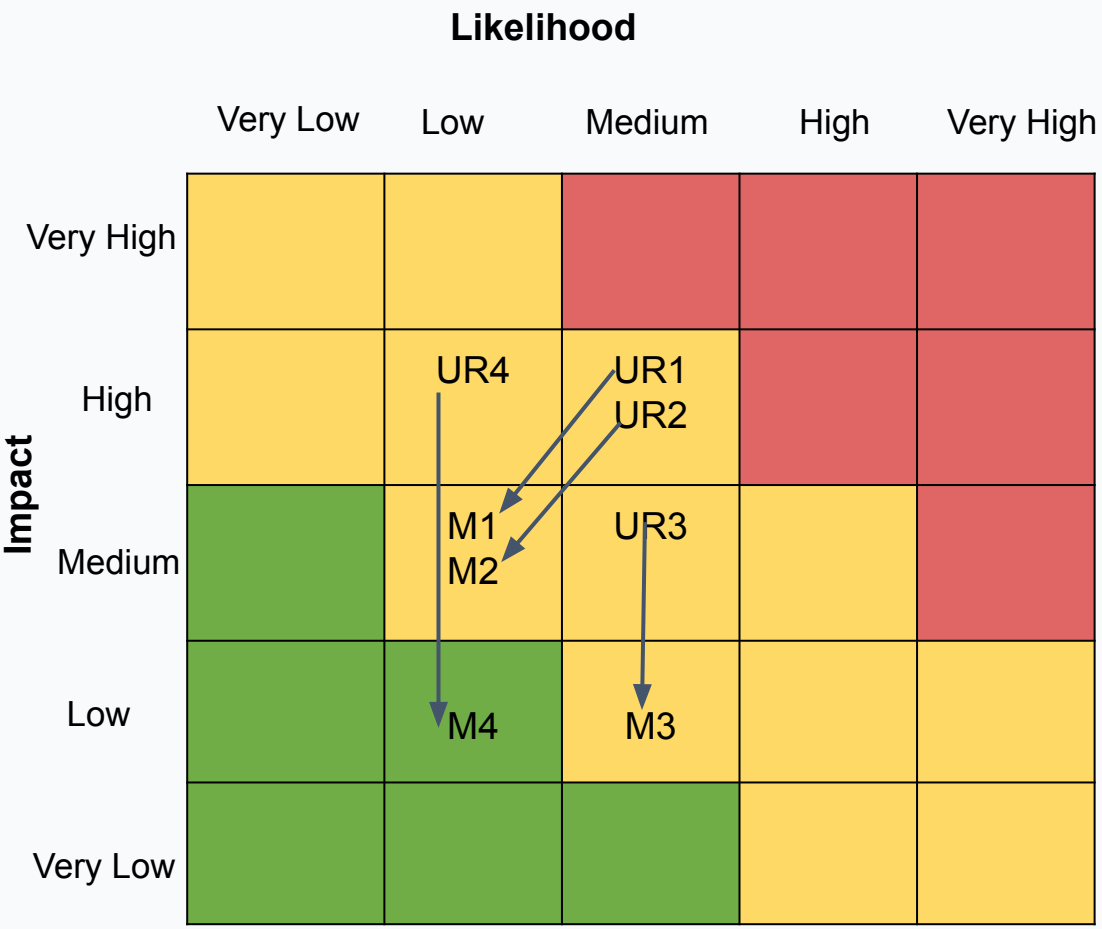






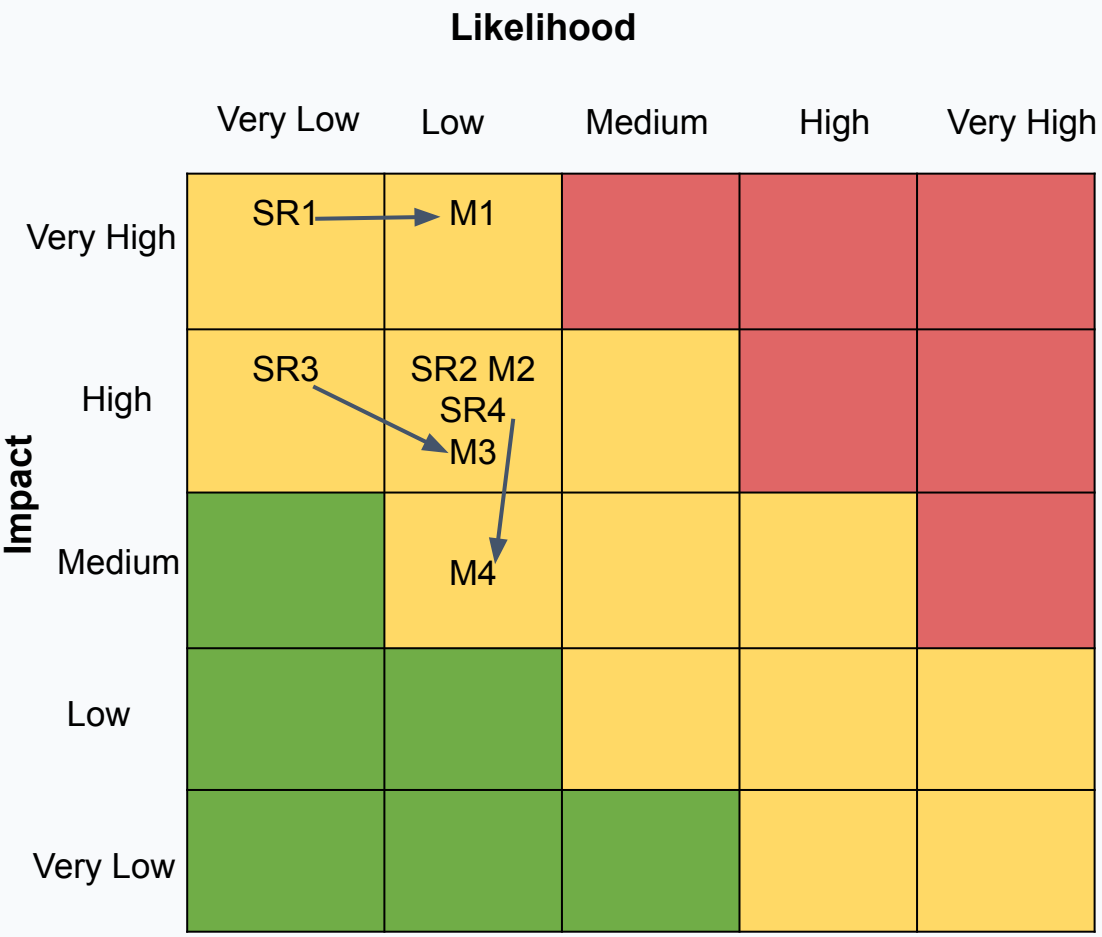


User Risk Mitigation Matrix



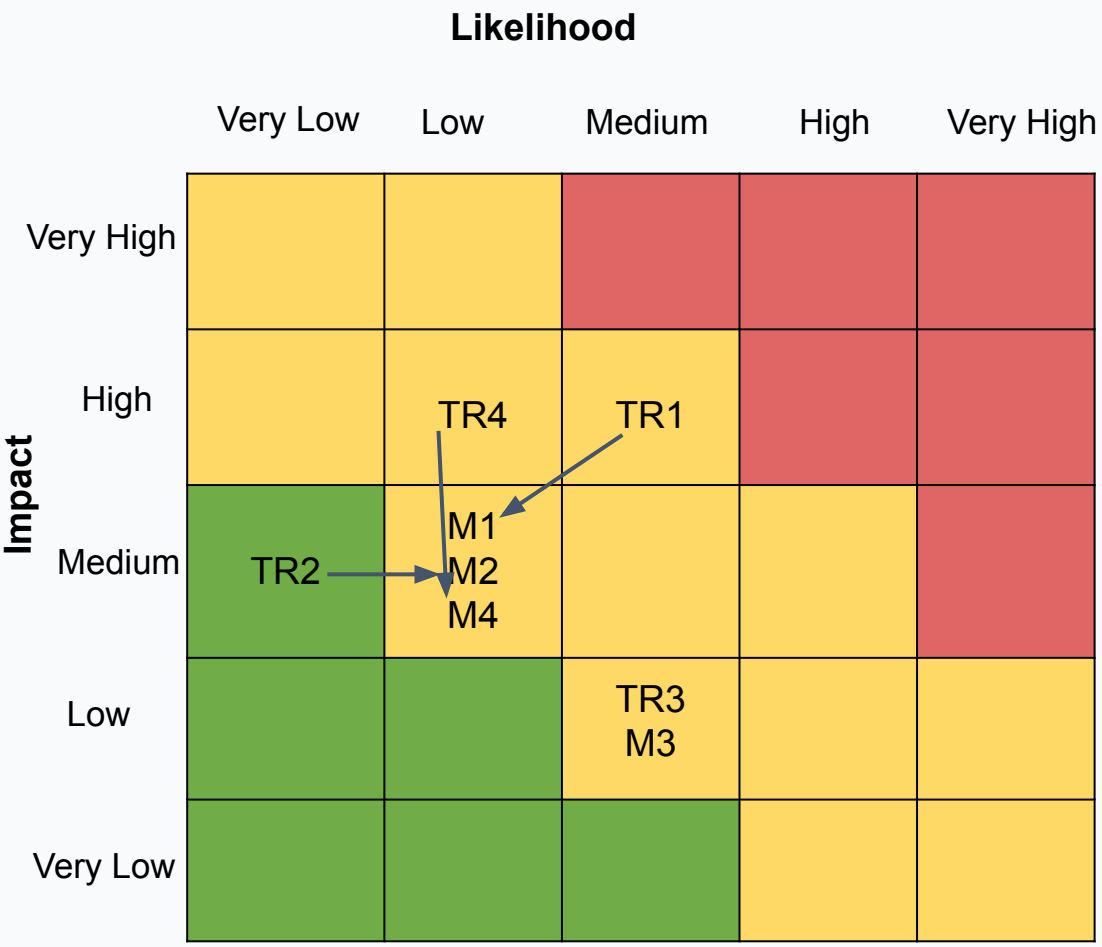
Risk ID	Description	Potential Impact / Likelihood	Technical & Operational Mitigation Strategy
R1 → M1 U1	Therapists enter incorrect assessment scores or incomplete patient information	**HIGH / MEDIUM**	Input validation, required fields, and clear instructions for entering assessment information.
R2 → M2 U2	Therapists rely on automatically generated goals without reviewing their clinical accuracy.	**HIGH / MEDIUM**	Mandatory therapist review and approval before goals can be exported.
R3 → M3 U3	Therapists are hesitant to adopt a new documentation system due to unfamiliarity or existing workflow	**MEDIUM / MEDIUM**	Provide an intuitive interface, onboarding instructions, and user training.
R4 → M4 U4	Users expose their login credentials through weak passwords or phishing attempts	**HIGH / LOW**	Strong authentication, requirements, secure password storage, and account security guidance.

Security Risk Mitigation Matrix



Risk ID	Description	Potential Impact / Likelihood	Technical & Operational Mitigation Strategy
R1 → M1 S1	Leakage of Identifiable patient health information.	**CRITICAL / VERY LOW**	Zero Persistent PHI architecture; client-side scrubbing of direct identifiers; ephemeral session tokens; TLS 1.3 transmission
R2 → M2 S2	Unauthorized individuals gain access to therapist accounts or clinical information.	**HIGH / LOW**	Secure authentication, password hashing, role-based access controls, and automatic session expiration
R3 → M3 S3	Patient-related assessment information is intercepted during transmission between the browser and server	**HIGH / VERY LOW**	HTTPS/TLS 1.3 encryption; secure communication protocols; restricted server access.
R4 → M4 S4	Patient related information is unintentionally exposed through external language-processing services	**HIGH / LOW**	Reducing transmitted information; remove direct patient identifiers before processing; use approved third-party services with appropriate privacy protections

Technical Risk Mitigation Matrix



Risk ID	Description	Potential Impact / Likelihood	Technical & Operational Mitigation Strategy
R1 → M1 T1	Language model invents patient deficits or unrealistic milestones.	**HIGH / MEDIUM**	Strict prompt bounding anchored exclusively to entered test scores; rule engine cross-checks outputs; mandatory therapist sign-off before export.
R2 → M2 T2	Target clinic EHR Software does not support direct API integration or compatible formatting	**MEDIUM / VERY LOW**	Clipboard export that produces standardized text compatible with existing EHR systems.
R3 → M3 T3	Application becomes unavailable due to server failures, maintenance, or unexpected technical issues	**MEDIUM / LOW**	Monitoring, error logging, reliable hosting, and recovery procedures.
R4 → M4 T4	Incorrect calculations or interpretation of standardized assessment scores produce inaccurate clinical observations	**HIGH / LOW**	Validize scoring calculations against authorized assessment procedures and implement automated unit tests.

Competitive Analysis: Market Gaps & True Innovation

Features	OT GoalAssist	SPRY Health	Pearson Q-Global	WebPT	ClinicSource	SimplePractice
Drafts C.O.A.S.T. Goals From Assessment Results and Patient Priorities	✔	✘	✘	✘	✘	✘
Checks Each Goal For All C.O.A.S.T. Elements	✔	✘	✘	✘	✘	✘
Omission Guardrails	✔	✔	✔	◐, Limited AI Flagging	✔	◐, Vendor Doesn't Mention
General AI Scribe	✘	✔	✔	✘	✘	◐, Higher Priced Tiers Only
Low Cost?	TBD	✘	✘	✘	✘	✘
Generic Goal Generation	✔	✔	✔	✔	✔	✔

Core Differentiator: Existing tools either offer rigid drop-downs or unconstrained text. OT GoalAssist is the first tool to directly translate standardized evaluation numbers into validated COAST goals with clinical omission guardrails.
"Rows 1 & 2: Data not found on vendor sites as of Nov. 8th."

Yes ✔ | Partial ◐ | No ✘

Clinical & Technical Glossary

Acronym / Clinical Term	Clinical Definition & Operational Context
ADL (Activities of Daily Living)	Fundamental self-care tasks required for daily independent living (dressing, eating, bathing, functional mobility).
AOTA	American Occupational Therapy Association; the primary national professional association representing occupational therapy clinicians.
BOT-2	Bruininks-Oseretsky Test of Motor Proficiency (2nd ed.); standardized clinical assessment battery measuring fine and gross motor skills.
COAST	Clinical goal documentation framework: Client, Occupation, Assist Level, Specific Conditions, and Time Frame.
COTA	Certified Occupational Therapy Assistant; works under registered occupational therapist supervision to deliver clinical interventions.
EHR	Electronic Health Record software platform utilized by clinical facilities to store patient charts and billing encounters.
MMT	Manual Muscle Testing; standardized 0-to-5 clinical scale used by therapists to grade patient muscular strength.
OTR	Occupational Therapist, Registered; licensed clinician qualified to conduct diagnostic evaluations and sign care plans.
PDMS-2	Peabody Developmental Motor Scales (2nd ed.); standardized assessment measuring early childhood gross and fine motor abilities.
POC (Plan of Care)	Formal clinical treatment roadmap detailing patient diagnoses, frequency, duration, and measurable functional goals.

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